RISHI RAJ

B. Tech: BIT Mesra | **E-Mail ID:** akrishiraj786@gmail.com | **Location:** Mumbai, Maharashtra, India | **Phone No:** +91-7050939324

SUMMARY

Recent graduate and aspiring Quantitative Researcher with strong academic foundation and experience in data science, mathematics, basic finance, Python, ML, SQL & statistics through projects and internships. Eager to contribute to Organization's quantitative research and trading initiatives.

EDUCATION

Degree	Board/University	Institute	%age/CGPA	Year
B. Tech (Mechanical Engineering)	BIT Mesra	Birla Institute of Technology, Mesra	**8.72 CGPA**	2021 - 2025
Class XII (Intermediate- STEM)	CBSE	D.A.V Public School	**90.0 %**	2018 - 2020
Class X (Matriculation)	ICSE	Tarapore School	**87.0 %**	2018

KEY SKILLS

Technical Skills

- **Statistical Programming Language:** C++, JAVA, Python
- **Database Management Tools:** SQL, MySQL, PostgreSQL, SQLite
- **Cloud & Automation Technologies:** AWS, Apache Airflow
- **Data Analytics & Visualization:** MS-Excel, MS-Power BI, Tableau, MS-PowerPoint, Google Sheet, Pivot Tables
- **Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, PyTorch, Tensorflow, Keras
- **Basic Quants & Finance:** Probability, Statistics, Quant Modelling, Risk Management, Basic Finance, Quant Finance

Soft Skills

Communication, Problem-Solving, Analytical Thinking, Team Player, Adaptability, Attention to Detail, Stakeholder Collaboration, Self-Motivated, Active Listener

INTERNSHIPS

Data Science Intern | **Tata Steel** | **May 2024-June 2024, Jamshedpur**

Responsibilities & Achievement

Project: *Increase in the reliability of the braking pinch roll assembly working inside R&D Dept. of New Bar Mill, TSL.*

- Gained valuable experience working in one of the India's leading R&D department and researched on various data science models that are being used for performance improvement of the braking pinch roll assembly working inside New Bar Mill,
- Achieved significant cost savings and elevated product quality through precise braking by developing and implementing data science model using raw data from machines and gained experience in data cleaning, pre-processing and EDA.

AI & Data Science Researcher | **BIT MESRA** | **Aug 2024-April 2025, Ranchi**

Responsibilities & Achievement

Project: *Optimization Studies on the Savonius Hydrokinetic Turbine Blade Design for Enhanced Torque Performance.*

- Implemented Genetic Algorithm and Neural Network (ANN) in CFD based to optimize blade redesign parameters leading to higher efficiency at different tip speed ratio (TSR) to calculate torque output coefficient (C_T) and power coefficient (C_p).
- Maximum efficiency was achieved by identifying the global maxima on the (C_T vs. TSR) and (C_P vs. TSR) plot, enhanced power coefficients via Tableau insights automation of design optimization techniques using AI computing-based methods

PROJECTS

Falcon 9 First Stage Landing Prediction | *IBM Cert.* | Python, NumPy, Pandas, Matplotlib, Seaborn, SQL, Scikit-learn, Plotly Dash, Folium, Request | *Link*

- Collected and cleaned Falcon 9 launch data using APIs, web scraping, SQL & Pandas to perform the exploratory data analysis to uncover launch trends.
- Applied predictive modeling on Falcon 9 launch time-series data and tested landing prediction robustness using cross-validation & feature engineering.
- Trained ML models (SVM, Decision Trees, Logistic) with GridSearchCV to predict first stage landing success, aiding mission planning and cost efficiency.

Time Series Forecasting of Equity Markets Using GRU & PyTorch | Python, PyTorch, GRU, Pandas, Matplotlib, Seaborn, Plotly | *Link*

- Forecasted stock prices of Google, Amazon, Microsoft & IBM (2006–2018) using GRU models to capture trends, volatility, and seasonal market behavior.
- Applied lag-based time series features & RMSE-driven model evaluation to optimize forecasting accuracy & generate reliable directional trading signals.
- Visualized predictive outputs & market shifts using Plotly, supporting backtest diagnostics & interpretability for financial modeling & strategy evaluation.

Natural Language Database Agent | *DeepLearning.AI Certificate* | Python, Azure OpenAI, LangChain, SQL, Assistants API, RAG, Function Calling | *Link*

- Developed a natural language agent using LangChain and Azure OpenAI to convert user prompts into secure SQL queries through function calling.
- Integrated Retrieval-Augmented Generation (RAG) to fetch relevant database and CSV context for accurate real-time query execution.
- Delivered a reusable AI-powered interface that minimized manual SQL use and enabled non-technical users to access structured data effortlessly.

Reddit ETL Data Pipeline Engineering | Python, AWS, Apache Airflow, PostgreSQL, AWS S3, AWS Glue, Amazon Athena, Amazon Redshift | *Link*

- Built a scalable ETL pipeline to extract structured signal data for downstream modeling, supporting signal generation workflows and historical analysis.
- Designed cloud-based data architecture on AWS S3 and schema cataloging using AWS Glue, enabling seamless querying through Amazon Athena.
- Automated data ingestion into Redshift and applied SQL-based transformations to support advanced analytics, reporting, and large-scale exploration.

POSITION OF RESPONSIBILITY

Aerospace Society, BIT MESRA

Management Head: *Responsible for the completion of organization projects and events.*

- Successfully organized the quiz, seminar and aircraft model making competition during the Techfest at BIT Mesra
- Designed and manufactured an autonomous UAV with a unique design go carry out suicide bombing missions.

Student Mentor

Workshop Organizer: *Organized different workshop (e.g., AutoCAD, SOLIDWORKS, CFD, ANSYS) workshops in CAD Lab.*
Teaching Assistant: *Responsible for delivering lectures to all first and second-year students who participated in workshop.*

CERTIFICATIONS & LICENSES

IBM Data Science Professional Certificate | *IBM via Coursera*

- Gained hands-on experience in data collection, cleaning, analysis, visualization (including Plotly Dash), and machine learning model development.
- Completed that associated project Falcon 9 First-Stage Landing Prediction, applying end-to-end data science techniques to real-world SpaceX data.

Machine Learning Specialization | *Stanford University & DeepLearning.AI via Coursera*

- Mastered supervised learning (linear and logistic regression), neural networks (TensorFlow), decision trees, ensemble methods, unsupervised learning (clustering, anomaly detection), recommender systems, and deep Q-learning reinforcement learning neural network to simulate lunar lander.
- Applied supervised & reinforcement learning, decision trees, and neural networks for predictive modeling on structure datasets with time-dependencies.

Preparatory Certificate in Finance and Financial Markets | *Corporate Finance Institute(CFI)*

- Gained practical understanding of capital markets, quant finance, financial institutions, accounting fundamentals, risk management & basic finance.
- Completed projects on investment risk modeling, stock return comparison using Google Sheets & portfolio diversification via correlation matrix analysis.

SCHOLASTIC ACHIEVEMENT

- Among the top 3%ile (Out of 1.6 Mn+) candidate in JEE Mains 2021 & top 4%ile (out of 0.24 Mn+) candidate in JEE Advanced 2021 & top 3% in BITSAT.
- Awarded a Merit Certification on the School Annual Appreciation Day in 2018, recognizing exemplary academic performance in the ICSE 2018.
- Received a Merit Certification on the School Annual Appreciation Day in 2020, recognizing exemplary academic performance in CBSE 2020.
- Selected as one of five members as Teaching Assistant for CAD Design & Modelling workshop to aid the instructor in delivering lectures & tutorials.

EXTRA CURRICULAR ACTIVITIES & INTEREST

- Participated in inter-hostel cricket tournament of BIT Mesra, core team member, all-rounder, served as vice-captain of Hostel-6 cricket team.
- Cricket, sports, swimming, culinary, dramatics and dance enthusiastic, and avid follower of Hollywood sci-fi, action and crime thriller films.